AI-Based FinOps Framework for MSME Cloud Cost Optimization using Predictive Analysis

1. Abstract

Cloud computing has become an essential technology for Micro, Small, and Medium Enterprises (MSMEs) due to its scalability, flexibility, and cost-effectiveness. However, many MSMEs struggle to efficiently manage cloud expenses because of unpredictable workloads, idle resources, and limited visibility into future cloud costs. Traditional cloud cost management tools mainly provide historical billing information and lack intelligent prediction and optimization capabilities.

This project proposes an **AI-Based FinOps Framework for MSME Cloud Cost Optimization using Predictive Analysis**. The framework utilizes Artificial Intelligence and Machine Learning techniques to analyze historical cloud usage data, predict future cloud expenditure, and identify opportunities for cost optimization. The system recommends actions such as resource right-sizing, identifying idle resources, storage optimization, and budget alerts. AWS cloud services including Amazon EC2, Amazon S3, Amazon RDS, AWS Lambda, Amazon SageMaker, Amazon CloudWatch, Amazon SNS, and AWS IAM are integrated to provide a scalable, secure, and automated cloud architecture. The proposed framework enables MSMEs to make informed financial decisions, reduce unnecessary cloud spending, improve resource utilization, and achieve efficient cloud operations through predictive analytics and intelligent recommendations.

2. Problem Statement

Many MSMEs are adopting cloud platforms such as AWS to host applications and store data. However, managing cloud costs is difficult due to dynamic pricing, over-provisioned resources, unused services, and unpredictable workloads. Existing cloud monitoring tools mainly provide historical billing information but do not accurately predict future expenses or recommend optimization strategies. Therefore, there is a need for an AI-powered FinOps framework that can forecast cloud costs, identify inefficient resource utilization, and provide intelligent recommendations to reduce cloud expenditure while maintaining performance and scalability.

3. Objectives

Develop an AI-based FinOps framework for cloud cost optimization.
Predict future cloud expenditure using machine learning models.
Detect underutilized cloud resources and unnecessary expenses.
Generate intelligent recommendations to reduce cloud costs.
Build an interactive dashboard for cloud cost monitoring and visualization.
Integrate AWS cloud services for secure, scalable, and reliable deployment.

4. Literature Survey

Paper	Method	Dataset	Advantages	Limitations	Research Gap
Towards Proactive Risk-Aware Cloud Cost Optimization Leveraging Transient Resources (IEEE, 2023)	Risk-aware optimization	Amazon EC2 Spot Instances	Reduces cloud cost while balancing risk	Focuses only on spot instances	Needs AI-based forecasting for MSMEs
Model-based Cloud Service Deployment Optimisation Method (Springer, 2023)	Model-based optimization	Cloud deployment models	Optimizes deployment before launch	Limited to deployment stage	Lacks real-time monitoring
Technical Note—Cloud Cost Optimization: Model, Bounds, and Asymptotics (Operations Research, 2024)	Mathematical optimization	AWS pricing data	Strong theoretical optimization	Complex implementation	Needs practical AI integration
Cost Modelling and Optimisation for Cloud: A Graph-based Approach (Springer, 2024)	Graph-based optimization	Multi-cloud deployment scenarios	Supports cost, QoS and scalability	High computational complexity	Limited MSME focus
Cloud PricingOps: A Decision	Decision Support System	Cloud pricing dataset	Supports FinOps decision-maki	No automated recommendat	AI-driven optimization required

Support Framework (MDPI, 2024)			ng	ions	
RAP-Optimizer: Resource-Aware Predictive Model (MDPI, 2024)	Deep Neural Network + Simulated Annealing	AIaaS application workload	High prediction accuracy and cost reduction	AIaaS-specific	General cloud workload support needed
Cloud Storage Cost: A Taxonomy and Survey (Springer, 2024)	Taxonomy & survey	Cloud storage pricing	Comprehensive storage cost analysis	Survey only	Predictive optimization absent
FinOps-Driven Strategies for Large-Scale Cloud Cost Optimization (2024)	FinOps framework	Enterprise cloud environments	Practical governance approach	Limited ML integration	Needs predictive analytics
FinOps at Scale: Reducing National Cloud Waste Through Predictive Optimization (2023)	Predictive FinOps	Multi-cloud workloads	Multi-cloud governance	Large-scale focus	MSME-specific framework needed
Cost and Performance Optimization in Cloud Computing (Procedia Computer Science, 2026)	Mixed-Integer Linear Programming + Predictive Methods	Cloud infrastructure workloads	Balances performance and cost	Early-stage evaluation	Real AWS deployment required
AI-Based Cloud Cost Prediction Framework	Machine Learning	Cloud billing data	Accurate cost forecasting	Requires large datasets	Improve explainability

(Related recent work)					
Intelligent Resource Allocation for Cloud Computing	Predictive resource allocation	Virtual machine usage	Better utilization	High computation	Integrate FinOps metrics
Cloud Resource Optimization Using Machine Learning	Regression models	Cloud utilization logs	Dynamic optimization	Limited cloud provider support	Multi-cloud compatibility
Predictive Analytics for Cloud Cost Management	Time-series forecasting	Historical billing records	Future cost estimation	Sensitive to data quality	Add anomaly detection
AI-Driven FinOps Framework for Cloud Optimization	Hybrid AI model	AWS billing and monitoring data	Combines prediction with recommendations	Prototype stage	Real-time production deployment

5. Research Gap Analysis

Existing Methods

Recent research has proposed several cloud cost optimization techniques, including risk-aware optimization, graph-based cloud resource modeling, predictive resource allocation, FinOps decision-support systems, and machine learning models for cloud cost forecasting. These approaches improve cloud resource utilization and reduce operational expenditure using optimization algorithms and predictive analytics.

Advantages

- Improves cloud resource utilization
- Reduces operational cloud expenditure
- Supports predictive cloud cost estimation
- Enhances decision making using AI
- Improves scalability and resource allocation
- Encourages FinOps best practices

Limitations

- Mostly designed for enterprise-scale organizations.
- Limited focus on MSMEs.
- Lack of real-time anomaly detection.
- Many solutions optimize only one aspect (storage, compute, or deployment).
- Limited integration of prediction, optimization, and visualization into one platform.
- Complex deployment and configuration.

Research Gap

Although several cloud cost optimization techniques exist, very few provide an end-to-end AI-powered FinOps framework specifically designed for MSMEs. Existing systems generally focus on cloud monitoring or cost optimization independently, while neglecting predictive analytics, intelligent recommendations, anomaly detection, and interactive visualization. There is a need for a unified solution that combines machine learning, AWS cloud services, and FinOps practices into a single scalable framework.

Proposed Improvement

The proposed framework integrates predictive machine learning models, AWS cloud services, automated cloud cost analysis, anomaly detection, and interactive dashboards into a unified FinOps platform. It provides intelligent cost optimization recommendations tailored for MSMEs, enabling organizations to proactively reduce cloud expenditure while maintaining application performance and scalability.

6. Novelty Summary

The proposed AI-Based FinOps Framework introduces a unified cloud cost optimization platform specifically designed for Micro, Small, and Medium Enterprises (MSMEs). Unlike existing cloud monitoring solutions that primarily provide historical billing information, the proposed framework predicts future cloud expenditure using machine learning algorithms and generates intelligent optimization recommendations.

The framework integrates AWS services including Amazon EC2, Amazon S3, Amazon RDS, AWS Lambda, Amazon SageMaker, Amazon CloudWatch, Amazon SNS, and AWS IAM into a secure and scalable cloud architecture. It also incorporates anomaly detection, automated budget alerts, and interactive dashboards to improve financial visibility and operational efficiency.

Key Novel Features

- AI-based cloud cost prediction.
- Automated cloud resource optimization recommendations.
- Real-time anomaly detection.
- Interactive dashboard for cloud expenditure analysis.
- AWS-native scalable architecture.
- FinOps practices specifically adapted for MSMEs.
- Predictive budget alerts and reporting.

7. Proposed AWS Architecture

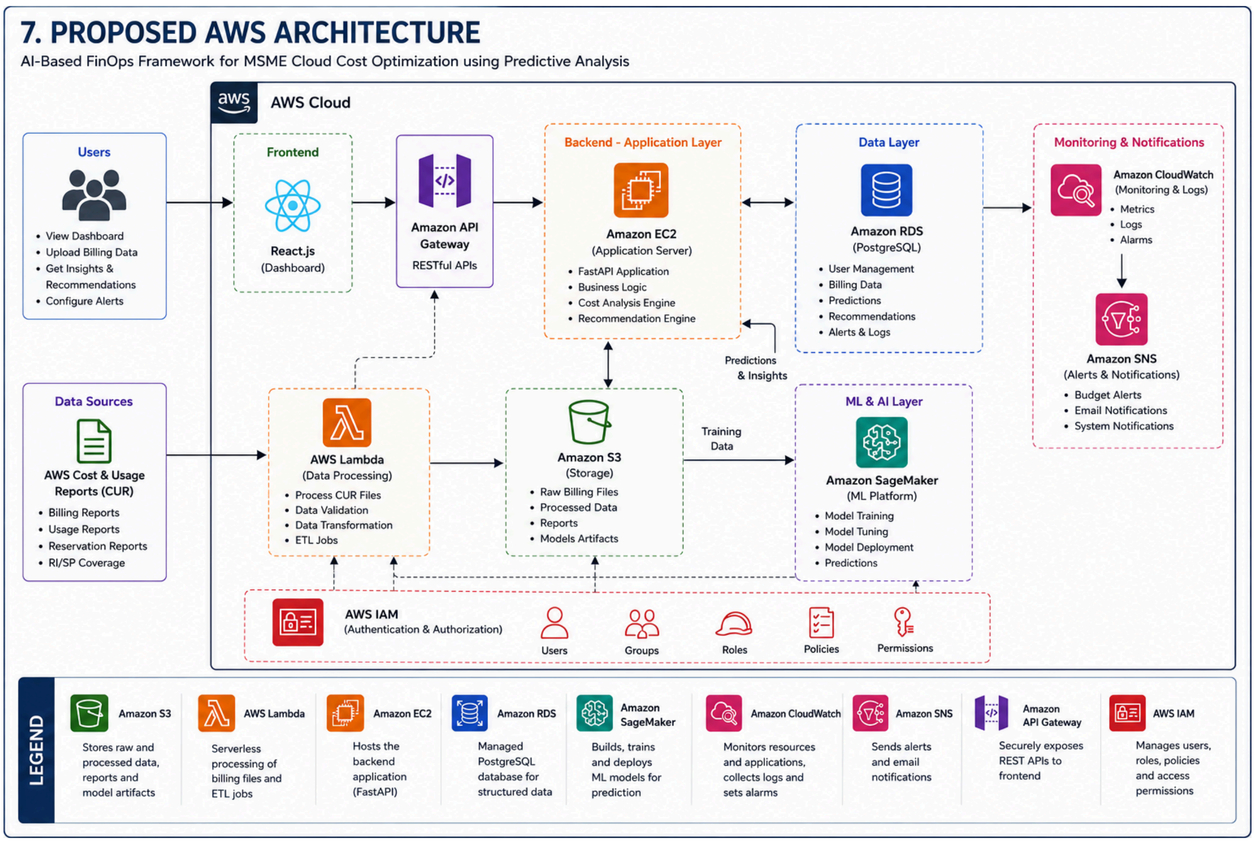

Figure 1. Proposed AWS Architecture of the AI-Based FinOps Framework.

8. Complete System Architecture

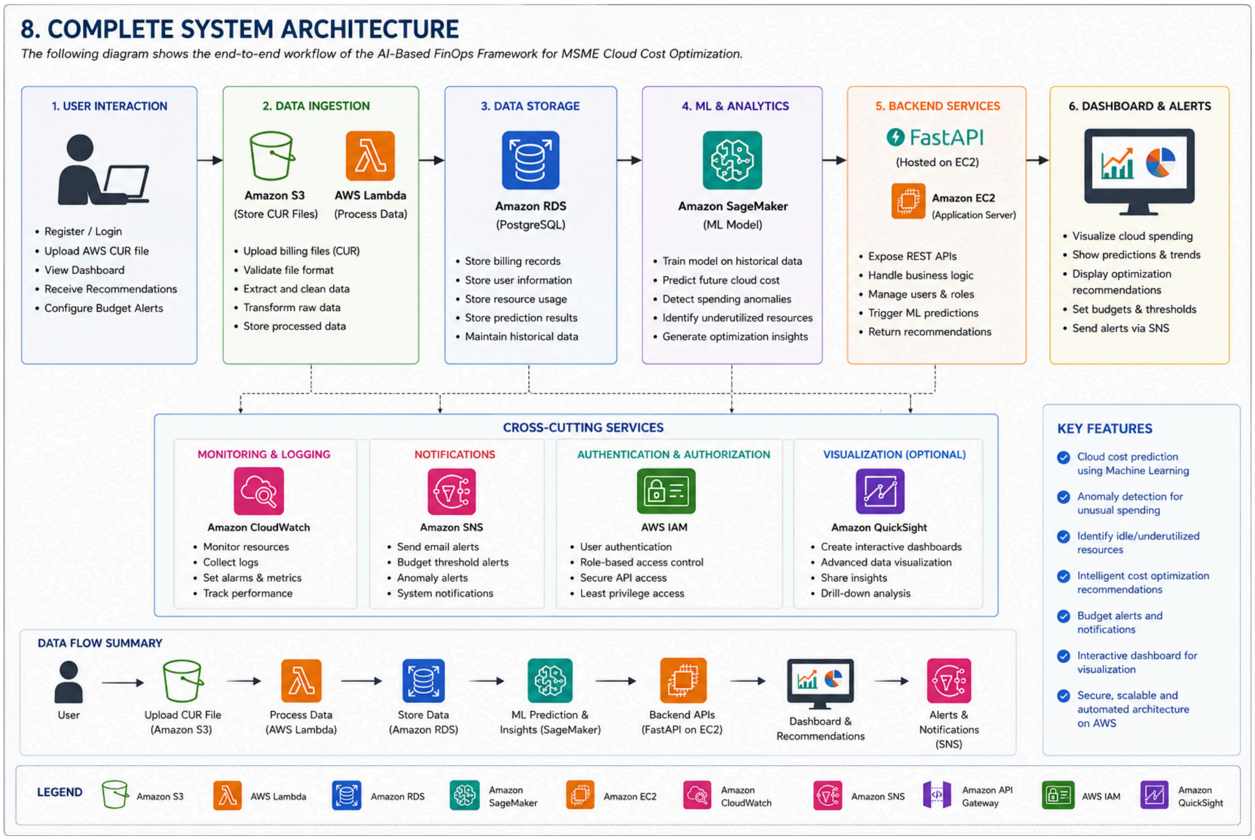

Figure 2. Complete System Workflow of the AI-Based FinOps Framework.

9. Dataset Details

Field	Details
Dataset Name	AWS Cost and Usage Reports (CUR)
Source	Amazon Web Services
URL	https://docs.aws.amazon.com/cur/
Dataset Size	Varies depending on AWS account usage
Number of Records	Dynamic (billing records generated daily)
Number of Features	20+ billing attributes
Data Type	CSV

Features Used	Service Name, Region, Usage Type, Usage Hours, Storage, Cost, Billing Period, Resource ID
License	AWS Customer Billing Data
Purpose	Train ML model for cloud cost prediction and optimization
Preprocessing	Remove missing values, normalize numerical data, encode categorical variables, feature selection

10. AWS Services Planning

AWS Service	Purpose
Amazon EC2	Host FastAPI backend
Amazon S3	Store billing datasets, reports, and ML artifacts
Amazon RDS	Store user accounts, billing history, predictions
AWS Lambda	Process uploaded billing files and ETL tasks
Amazon SageMaker	Train and deploy machine learning models
Amazon API Gateway	Secure REST API communication
AWS IAM	Authentication and authorization
Amazon CloudWatch	Monitoring, metrics, and logging
Amazon SNS	Budget alerts and notifications
Amazon QuickSight	Interactive analytics dashboards

11. Technology Stack

Category	Technology
Frontend	React.js, Tailwind CSS, Chart.js
Backend	FastAPI, Python

Database	PostgreSQL
AI/ML	Scikit-learn, XGBoost, Pandas, NumPy
Cloud	AWS EC2, S3, RDS, Lambda, SageMaker
APIs	REST API, Amazon API Gateway
Monitoring	Amazon CloudWatch
Notifications	Amazon SNS
Authentication	AWS IAM

12. Work Distribution

Student	**Responsibilities**
Ridhima Singh	Frontend development, Dashboard UI, Documentation
Rahul Ganesan	Backend development, Database integration, AWS cloud services
Akhilesh Reddy	AI/ML model development, Dataset preprocessing, Prediction engine, Testing